
User Manual

for MPC560XB WDG Driver

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Rev 1.2





Contents

Section number	Title	Page
Chapter 1		
Revision History		
Chapter 2		
Introduction		
2.1	Supported Derivatives.....	9
2.2	Overview.....	10
2.3	About this Manual.....	10
2.4	Acronyms and Definitions.....	11
2.5	Reference List.....	11
Chapter 3		
Driver		
3.1	Requirements.....	13
3.2	Driver Design Summary.....	13
3.3	Deviation from Requirements.....	14
3.4	Software specification.....	15
3.4.1	Define Reference.....	16
3.4.2	Macros Reference.....	16
3.4.2.1	Macro SWT_CO.....	16
3.4.2.2	Macro SWT_CR.....	16
3.4.2.3	Macro SWT_GET_KEYED_SERVICE_SEQ1.....	16
3.4.2.4	Macro SWT_IR.....	16
3.4.2.5	Macro SWT_SK.....	16
3.4.2.6	Macro SWT_SR.....	17
3.4.2.7	Macro SWT_TO.....	17
3.4.2.8	Macro SWT_WN.....	17
3.4.3	Enum Reference.....	17
3.4.3.1	Enumeration Wdg_StatusType.....	17

Section number	Title	Page
3.4.4	Function Reference.....	18
3.4.4.1	Function Wdg_ChannelGetVersionInfo.....	18
3.4.4.2	Function Wdg_ChannelInit.....	18
3.4.4.3	Function Wdg_ChannelSetMode.....	19
3.4.4.4	Function Wdg_ChannelSetTriggerCondition.....	20
3.4.5	Structs Reference.....	21
3.4.6	Types Reference.....	21
3.4.7	Variables Reference.....	21
3.5	Symbolic Names DISCLAIMER.....	21
3.6	Hardware Errata details for MPC560XB.....	21

Chapter 4 Tresos Configuration Plug-in

4.1	Configuration elements of Wdg.....	23
4.2	Form IMPLEMENTATION_CONFIG_VARIANT.....	23
4.3	Form NonAutosar.....	24
4.3.1	WdgDisableDemReportErrorStatus (NonAutosar).....	24
4.4	Form WdgDemEventParameterRefs.....	25
4.4.1	WDG_E_DISABLE_REJECTED (WdgDemEventParameterRefs).....	25
4.4.2	WDG_E_MODE_FAILED (WdgDemEventParameterRefs).....	25
4.4.3	WDG_E_MISS_TRIGGER (WdgDemEventParameterRefs).....	25
4.5	Form WdgGeneral.....	26
4.5.1	WdgDevErrorDetect (WdgGeneral).....	26
4.5.2	WdgDisableAllowed (WdgGeneral).....	27
4.5.3	WdgMasterAccessProtection (WdgGeneral).....	27
4.5.4	WdgIndex (WdgGeneral).....	28
4.5.5	WdgInitialTimeout (WdgGeneral).....	28
4.5.6	WdgMaxTimeout (WdgGeneral).....	29
4.5.7	WdgRunArea (WdgGeneral).....	29
4.5.8	WdgCallbackNotification0 (WdgGeneral).....	30

Section number	Title	Page
4.5.9	WdgTriggerLocation (WdgGeneral).....	30
4.5.10	WdgVersionInfoApi (WdgGeneral).....	31
4.6	Form WdgPublishedInformation.....	31
4.6.1	WdgTriggerMode (WdgPublishedInformation).....	31
4.7	Form CommonPublishedInformation.....	32
4.7.1	ArReleaseMajorVersion (CommonPublishedInformation).....	32
4.7.2	ArReleaseMinorVersion (CommonPublishedInformation).....	33
4.7.3	ArReleaseRevisionVersion (CommonPublishedInformation).....	33
4.7.4	ModuleId (CommonPublishedInformation).....	34
4.7.5	SwMajorVersion (CommonPublishedInformation).....	34
4.7.6	SwMinorVersion (CommonPublishedInformation).....	35
4.7.7	SwPatchVersion (CommonPublishedInformation).....	35
4.7.8	VendorApiInfix (CommonPublishedInformation).....	36
4.7.9	VendorId (CommonPublishedInformation).....	36
4.8	Form WdgSettingsConfig.....	37
4.8.1	WdgInstance (WdgSettingsConfig).....	38
4.8.2	WdgDefaultMode (WdgSettingsConfig).....	38
4.8.3	WdgKeyedService (WdgSettingsConfig).....	39
4.8.4	WdgServiceKeyValue (WdgSettingsConfig).....	39
4.8.5	WdgExternalTriggerCounterRef (WdgSettingsConfig).....	40
4.8.6	Form WdgSettingsFast.....	40
4.8.6.1	WdgReadOnly (WdgSettingsFast).....	41
4.8.6.2	WdgHardLock (WdgSettingsFast).....	41
4.8.6.3	WdgWindowMode (WdgSettingsFast).....	42
4.8.6.4	WdgRunsInStopMode (WdgSettingsFast).....	42
4.8.6.5	WdgRunsInDebugMode (WdgSettingsFast).....	43
4.8.6.6	WdgOperationMode (WdgSettingsFast).....	43
4.8.6.7	WdgResetOnInvalidAccess (WdgSettingsFast).....	44
4.8.6.8	WdgClockSelection (WdgSettingsFast).....	44

Section number	Title	Page
4.8.6.9	WdgTimeoutPeriod (WdgSettingsFast).....	45
4.8.6.10	WdgWindowPeriod (WdgSettingsFast).....	45
4.8.7	Form WdgSettingsOff.....	46
4.8.7.1	WdgReadOnly (WdgSettingsOff).....	46
4.8.7.2	WdgHardLock (WdgSettingsOff).....	47
4.8.8	Form WdgSettingsSlow.....	47
4.8.8.1	WdgReadOnly (WdgSettingsSlow).....	48
4.8.8.2	WdgHardLock (WdgSettingsSlow).....	48
4.8.8.3	WdgWindowMode (WdgSettingsSlow).....	49
4.8.8.4	WdgRunsInStopMode (WdgSettingsSlow).....	49
4.8.8.5	WdgRunsInDebugMode (WdgSettingsSlow).....	50
4.8.8.6	WdgOperationMode (WdgSettingsSlow).....	50
4.8.8.7	WdgResetOnInvalidAccess (WdgSettingsSlow).....	51
4.8.8.8	WdgClockSelection (WdgSettingsSlow).....	51
4.8.8.9	WdgTimeoutPeriod (WdgSettingsSlow).....	52
4.8.8.10	WdgWindowPeriod (WdgSettingsSlow).....	52

Chapter 1

Revision History

Table 1-1. Revision History

Revision	Date	Author	Description
1.0	30-Jan-2012	Salvatore Nappi	Initial release for MPC560XB
1.1	10-May-2013	Jayant Shekhar	Bolero ASR 4.0 RTM 1.0.0 Release
1.2	30-April-2014	Salvatore Nappi	Added multinstance reference and callback specification



Chapter 2

Introduction

This User Manual describes Freescale Semiconductor AUTOSAR Watchdog (WDG) for MPC560XB .

AUTOSAR WDG driver configuration parameters and deviations from the specification are described in WDG Driver chapter of this document. AUTOSAR WDG driver requirements and APIs are described in the AUTOSAR WDG driver software specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of Freescale Semiconductor .

Table 2-1. MPC560XB Derivatives

Freescale Semiconductor	mpc5601dxlh_lqfp64 mpc5601dxll_lqfp100 mpc5602dxlh_lqfp64 mpc5602dxll_lqfp100 mpc5602bxlh_lqfp64 mpc5602bxll_lqfp100 mpc5602bxlq_lqfp144 mpc5602cxlh_lqfp64 mpc5602cxll_lqfp100 mpc5603bxlh_lqfp64 mpc5603bxll_lqfp100 mpc5603bxlq_lqfp144 mpc5603cxlh_lqfp64 mpc5603cxll_lqfp100 mpc5604bxlh_lqfp64 mpc5604bxll_lqfp100 mpc5604bxlq_lqfp144 mpc5604bxmg_mapbga208 mpc5604cxlh_lqfp64 mpc5604cxll_lqfp100 mpc5605bxll_lqfp100 mpc5605bxlq_lqfp144 mpc5605bxlu_lqfp176 mpc5606bxlq_lqfp144
-------------------------	--

Table 2-1. MPC560XB Derivatives

	mpc5606bxlu_lqfp176 mpc5607bxlu_lqfp176 mpc5607bxmg_mapbga208
--	---

All of the above microcontroller devices are collectively named as MPC560XB .

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About this Manual

This Technical Reference employs the following typographical conventions:

Boldface type: Bold is used for important terms, notes and warnings.

Italic font: Italic typeface is used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

2.4 Acronyms and Definitions

Table 2-2. Acronyms and Definitions

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
ASM	Assembler
DEM	Diagnostic Event Manager
DET	Development Error Tracer
C/CPP	C and C++ Source Code
VLE	Variable Length Encoding
N/A	Not Applicable
MCU	Micro Controller Unit
PB Variant	Post Build Variant
PC Variant	Pre Compile Variant
LT Variant	Link Time Variant
OS	Operating System
ISR	Interrupt Service Routine

2.5 Reference List

Table 2-3. Reference List

#	Title	Version
1	AUTOSAR 4.0 Rev0003WDG Driver Software Specification Document.	2.5.0
2	MPC5607B Reference Manual	Rev. 7.2, May 2012
3	MPC5604B Reference Manual	Rev. 8.1, May 2012
4	5606B Reference Manual	Rev. 1, 12 Jul 2011
5	5602D Reference Manual	Rev. 4.1, May 2012
6	Bolero 1.5M cut 2.2 Errata Sheet Reference Manual	MPC5607B_1M03Y - Rev. 30Aug2012
7	Bolero 512k cut 2.3 Errata Sheet Reference Manual	MPC5604B_2M27V - Rev 17 MAY 2012
8	Bolero 256k cut 1.1 Errata Sheet Reference Manual	MPC5602D_1M18Y - Rev 17 MAY 2012

Chapter 3 Driver

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR 4.0 Rev0003WDG Driver Software Specification document (See Table [Reference List](#)).

3.2 Driver Design Summary

A Software Watchdog Timer (SWT) with programmable interrupt response is available in MPC560XB . The Software Watchdog Timer (SWT) is a peripheral module that can prevent system lockup in situations such as software getting trapped in a loop or if a bus transaction fails to terminate. When enabled, the SWT requires periodic execution of a watchdog servicing operation. The servicing operation resets the timer to a specified time-out period. If this servicing action does not occur before the timer expires the SWT generates an interrupt or hardware reset. The SWT can be configured to generate a reset or interrupt on an initial time-out, a reset is always generated on a second consecutive time-out.

In addition to these modes of operation, the watchdog timer also supports a windowed mode of operation. In this mode, the servicing of the watchdog timer must be performed in the last part of the timeout period defined by the window register. The window is open when the down counter is less than the value in the SWT_WN register. Outside of this window, service sequence writes are invalid accesses and generate a bus error or reset depending on the value of the SWT_CR.RIA. These timeout responses are configurable using the configuration parameter ‘WdgOperationMode’

The SWT has the following features:

- 32-bit time-out register to set the time-out period
- Oscillator clock for timer operation

Deviation from Requirements

- Programmable selection of window mode or regular servicing
- Programmable selection of reset or interrupt on an initial time-out
- Programmable selection of fixed or keyed servicing
- Master access protection

The SWT is started on exit of power-on phase (RGM phase 2) to monitor flash boot sequence phase. It is then reset during (RGM phase 3) and optionally enabled when platform reset is released depending on value of flash user option bit 31 (WATCHDOG_EN)

3.3 Deviation from Requirements

The driver deviates from the AUTOSAR WDG Driver software specification in some places. Table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out of scope for the WDG driver. Table [Table 3-1](#) provides Status column description.

Table 3-1. Deviations Status Column Description

Term	Definition
N/A	Not available
N/T	Not testable
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented
N/R	Unclear Requirement
N/V	Not Verifiable

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, or out of scope for the driver.

Table 3-2. Driver Deviations Table

Requirement	Status	Description	Notes
WDG013	N/A	Additional errors that are detected because of specific implementation and/or specific hardware properties shall be added in the module's implementation documentation. The classification and enumeration shall be compatible to the errors listed above [WDG010]	No additional errors are supported by hardware, hence no additional errors are implemented.
WDG034	N/A	General design rules: The start address of the watchdog trigger routine shall be statically configurable to a fixed memory location by the	Not supported by hardware

Table continues on the next page...

Table 3-2. Driver Deviations Table (continued)

Requirement	Status	Description	Notes
		user. The user needs to take care that Configured memory location is valid for the platform on which driver is being implemented on. This configuration parameter shall only be given if supported/needed by the hardware. Rationale: This allows the watchdog device to identify the correct trigger input if supported by the hardware.	
WDG040	N/A	If interrupts have to be disabled in order to ensure data consistency or correct functionality of this module (e.g. while switching the watchdog mode or during the watchdog trigger routine), this shall be done by using the corresponding BSW Scheduler functionality if possible	
WDG076	N/S	External watchdog driver To access the external watchdog hardware, the Wdg module shall use the functionality and API of the corresponding handler or driver, e.g. the SPI handler or DIO driver.	External module is customer dependant and is not yet developed.
WDG077	N/S	External watchdog driver - A Wdg module for an external watchdog shall satisfy the same functional requirements and offer the same functional scope as a Wdg module for an internal watchdog. Hence their respective APIs are semantically identical.	External module is customer dependant and is not yet developed.
WDG078	N/S	External watchdog driver - The Wdg module shall add all parameters required for accessing the external watchdog hardware, e.g. the used SPI channel or DIO port, to the module's published parameters and to the module's configuration parameters.	External module is customer dependant and is not yet developed.
WDG113_Conf	N/S	WdgExternalContainerRef container definition	External module is customer dependant and is not yet developed.
WDG118_Conf	N/A	Wdg_Configuration - Location (memory address) of the watchdog trigger routine. Dependency: Only relevant if provided by hardware and needed by the system.	Not supported by hardware.
WDG140	N/F	Wdg_SetTriggerCondition - This function shall also allow to set "0" as the time frame for triggering which will result in an (almost) immediate stop of the watchdog triggering and an (almost) instantaneous watchdog reset of the ECU.	Not supported by hardware. The reset only occurs after the SWT timer expires. A new call of Wdg_SetTriggerCondition should not be performed during this period.

3.4 Software specification

The following sections contains driver software specifications.

3.4.1 Define Reference

Constants supported by the WDG driver.

3.4.2 Macros Reference

Macros supported by the WDG driver.

3.4.2.1 Macro SWT_CO

This constant specifies the SWT Output Register.

Prototype: `SWT_CO(u8ModuleIdx)`

3.4.2.2 Macro SWT_CR

This constant specifies the SWT Control Register.

Prototype: `SWT_CR(u8ModuleIdx)`

3.4.2.3 Macro SWT_GET_KEYED_SERVICE_SEQ1

This constant specifies the value of the second part of the fixed service sequence which resets the timeout value = $((17UL * REG_READ32(SWT_SK)) + 3UL) \bmod 2^{16}$

Prototype: `SWT_GET_KEYED_SERVICE_SEQ1(u8ModuleIdx)`

3.4.2.4 Macro SWT_IR

This constant specifies the SWT Interrupt Register.

Prototype: `SWT_IR(u8ModuleIdx)`

3.4.2.5 Macro SWT_SK

This constant specifies the SWT Service Key Register.

Prototype: `SWT_SK(u8ModuleIdx)`

3.4.2.6 Macro SWT_SR

This constant specifies the SWT Service Register.

Prototype: `SWT_SR(u8ModuleIdx)`

3.4.2.7 Macro SWT_TO

This constant specifies the SWT Time-out Register.

Prototype: `SWT_TO(u8ModuleIdx)`

3.4.2.8 Macro SWT_WN

This constant specifies the SWT Window Register.

Prototype: `SWT_WN(u8ModuleIdx)`

3.4.3 Enum Reference

Enumeration of all constants supported by the WDG driver.

3.4.3.1 Enumeration Wdg_StatusType

This enumerated type will contain the watchdog driver's possible states.

Table 3-3. Enumeration Wdg_StatusType Values

Name	Initializer	Description
WDG_UNINIT	= 0x00	The watchdog driver is not uninitialized.
WDG_IDLE		= 0x01 The watchdog driver is currently idle, i.e not beeing triggered or mode changed.
WDG_BUSY		= 0x02 The watchdog driver is currently busy, i.e triggered or switchd between modes.

3.4.4 Function Reference

Functions supported by the WDG driver.

3.4.4.1 Function Wdg_ChannelGetVersionInfo

Returns the version information of the module.

Details:

The Wdg_GetVersionInfo function shall return the version information of this module. The version information includes:

- Module Id,
- Vendor Id,
- Vendor specific version numbers.

Pre: This ISR is only required if the WDG_VERSION_INFO_API has to be equal STD_ON.

Return: void.

Implements: DWDG03000

Prototype: void Wdg_ChannelGetVersionInfo(Std_VersionInfoType *versioninfo, const Wdg_IPW_InstanceType Wdg_Instance);

Table 3-4. Wdg_ChannelGetVersionInfo Arguments

Type	Name	Direction	Description
Std_VersionInfoType *	versioninfo	input, output	Pointer to where to store the version information of this module.
constWdg_IPW_InstanceType	Wdg_Instance	input	SWT instance.

3.4.4.2 Function Wdg_Channellnit

This function initializes the WDG module.

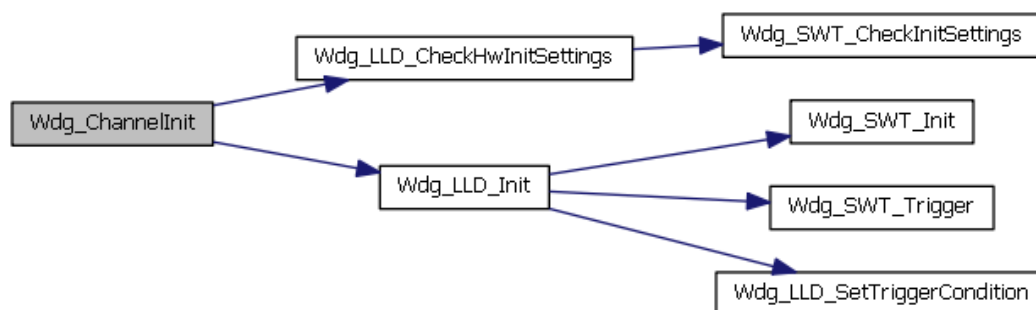


Figure 3-1. Function Wdg_ChannelInit References.

Details:

The `Wdg_Init` function shall initialize the Wdg module and the watchdog hardware, i.e. it shall set the default watchdog mode and timeout period as provided in the configuration set.

Return: void.

Implements: DWDG03001

Prototype: void Wdg_ChannelInit(const Wdg_ConfigType *ConfigPtr, const Wdg_IPW_InstanceType Wdg_Instance);

Table 3-5. Wdg_ChannelInit Arguments

Type	Name	Direction	Description
const Wdg_ConfigType *	ConfigPtr	input	Pointer to configuration set.
const Wdg_IPW_InstanceType	Wdg_Instance	input	SWT instance.

3.4.4.3 Function Wdg_ChannelSetMode

Switches the watchdog into the mode Mode.

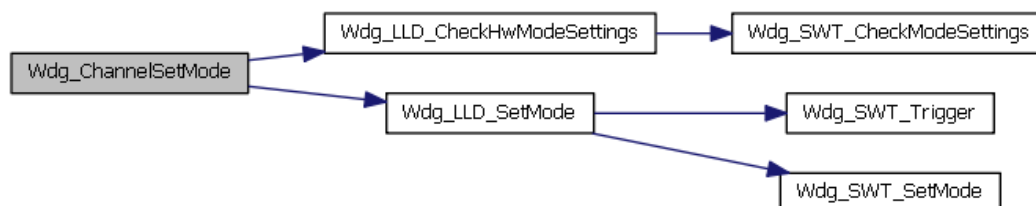


Figure 3-2. Function Wdg_ChannelSetMode References.

Details:

By choosing one of a limited number of statically configured settings (e.g. toggle or window watchdog, different timeout periods) the Wdg module and the watchdog hardware can be switched between the following three different watchdog modes using the `Wdg_SetMode` function:

- `WDGIF_OFF_MODE`,
- `WDGIF_SLOW_MODE`,
- `WDGIF_FAST_MODE`.

Return: `Std_ReturnType`.

Implements: `DWDG03002`

Prototype: `Std_ReturnType Wdg_ChannelSetMode(WdgIf_ModeType Mode, const Wdg_IPW_InstanceType Wdg_Instance);`

Table 3-6. Wdg_ChannelSetMode Arguments

Type	Name	Direction	Description
<code>WdgIf_ModeType</code>	<code>Mode</code>	input	One of the following statically configured modes: <code>WDGIF_OFF_MODE</code> , <code>WDGIF_SLOW_MODE</code> , <code>WDGIF_FAST_MODE</code> .
<code>constWdg_IPW_InstanceType</code>	<code>Wdg_Instance</code>	input	SWT instance.

Table 3-7. Wdg_ChannelSetMode Return Values

Name	Description
<code>E_OK</code>	Mode switch executed completely and successfully.
<code>E_NOT_OK</code>	The mode switch encountered errors.

3.4.4.4 Function `Wdg_ChannelSetTriggerCondition`

Reset the watchdog timeout counter according to the timeout value passed.



Figure 3-3. Function `Wdg_ChannelSetTriggerCondition` References.

Return: `void`.

Implements: DWDG03003

Prototype: void Wdg_ChannelSetTriggerCondition(uint16 Timeout, const Wdg_IPW_InstanceType Wdg_Instance);

Table 3-8. Wdg_ChannelSetTriggerCondition Arguments

Type	Name	Direction	Description
uint16	Timeout	input	Value (milliseconds) for setting the trigger counter.
constWdg_IPW_InstanceType	Wdg_Instance	input	SWT instance.

3.4.5 Structs Reference

Data structures supported by the WDG driver.

3.4.6 Types Reference

Types structures supported by the WDG driver.

3.4.7 Variables Reference

Variables supported by the WDG driver.

3.5 Symbolic Names DISCLAIMER

All containers having the symbolic name tag set as true in the Autosar schema will generate defines like:

```
#define <Container_Short_Name> <Container_ID>
```

For this reason it is forbidden to duplicate the name of such containers across the MCAL configuration, or to use names that may trigger other compile issues (e.g. match existing #ifdefs arguments).

3.6 Hardware Errata details for MPC560XB

While in STOP mode, if the SWT is configured to generate an interrupt and the system clock is disabled (`ME_STOP0_MC[SYSCLK] = 0xF`), a SWT interrupt event will not trigger an exit from STOP mode. Other internal or external wakeup events (RTC, API, WKUP pins) are not affected and will trigger a STOP exit independent of the `ME_STOP_MC[SYSCLK]` configuration.

For this reason when the SWT is in STOP mode the system clock should not be disabled.

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the WDG Driver. The most of the parameters are described below.

4.1 Configuration elements of Wdg

Included forms :

- IMPLEMENTATION_CONFIG_VARIANT
- NonAutosar
- WdgDemEventParameterRefs
- WdgGeneral
- WdgPublishedInformation
- CommonPublishedInformation
- WdgSettingsConfig

Table 4-1. Revision table

Revision	Date
4.1.0	2010-12-03

4.2 Form IMPLEMENTATION_CONFIG_VARIANT

Figure 4-1. Tresos Plugin snapshot for IMPLEMENTATION_CONFIG_VARIANT form.

Table 4-2. Attribute IMPLEMENTATION_CONFIG_VARIANT detailed description

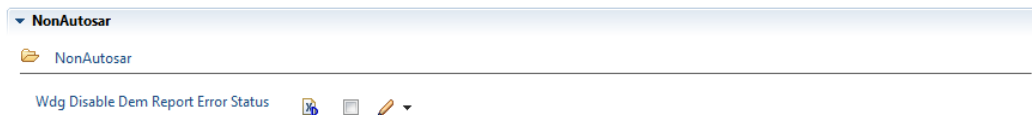
Property	Value
Label	Config Variant
Type	ENUMERATION
Default	VariantLinkTime
Range	VariantLinkTime VariantPostBuild VariantPreCompile

4.3 Form NonAutosar

NonAutosar

Autosar Requirements:

This container contains the global configuration parameters of the Non-Autosar Wdg driver. This container is a MultipleConfigurationContainer, i.e. this container and its sub-containers exist once per configuration set.

**Figure 4-2. Tresos Plugin snapshot for NonAutosar form.**

4.3.1 WdgDisableDemReportErrorStatus (NonAutosar)

Wdg Disable Dem Report Error Status

Switches the Diagnostic Error Reporting and Notification OFF.

Table 4-3. Attribute WdgDisableDemReportErrorStatus (NonAutosar) detailed description

Property	Value
Label	Wdg Disable Dem Report Error Status
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.4 Form WdgDemEventParameterRefs

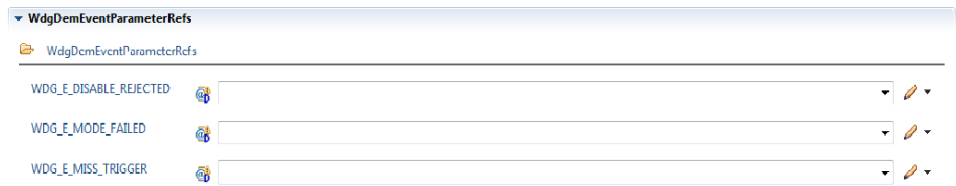


Figure 4-3. Tresos Plugin snapshot for WdgDemEventParameterRefs form.

4.4.1 WDG_E_DISABLE_REJECTED (WdgDemEventParameterRefs)

Table 4-4. Attribute WDG_E_DISABLE_REJECTED (WdgDemEventParameterRefs) detailed description

Property	Value
Type	SYMBOLIC-NAME-REFERENCE
Origin	AUTOSAR_ECUC
Enable	true

4.4.2 WDG_E_MODE_FAILED (WdgDemEventParameterRefs)

Table 4-5. Attribute WDG_E_MODE_FAILED (WdgDemEventParameterRefs) detailed description

Property	Value
Type	SYMBOLIC-NAME-REFERENCE
Origin	AUTOSAR_ECUC
Enable	true

4.4.3 WDG_E_MISS_TRIGGER (WdgDemEventParameterRefs)

Table 4-6. Attribute WDG_E_MISS_TRIGGER (WdgDemEventParameterRefs) detailed description

Property	Value
Type	SYMBOLIC-NAME-REFERENCE
Origin	Custom
Enable	true

4.5 Form WdgGeneral

WdgGeneral

All general parameters of the watchdog driver are collected here.

The screenshot shows the TRESOS Plugin configuration window for the WdgGeneral form. The window has a title bar with a dropdown menu set to 'WdgGeneral'. Below the title bar, there is a list of parameters and their values:

- Development Error Detection: ☒ (icon)
- WdgMasterAccessProtection: ALL_MAPx (dropdown menu)
- Wdg Index (0 -> 255): 0 (text field)
- Wdg Initial Timeout [s] (0 -> 65.535): 0.1 (text field)
- Wdg Max Timeout [s] (0 -> 65.535): 0.2 (text field)
- Wdg Run Area: ROM (dropdown menu)
- Wdg CallBack Notification: NULL_PTR (text field)
- Wdg Trigger Location: pointer (text field)
- Provide Version Info API: ☒ (icon)

Figure 4-4. TRESOS Plugin snapshot for WdgGeneral form.

4.5.1 WdgDevErrorDetect (WdgGeneral)

Wdg Development Error Detect

Compile switch to enable / disable development error detection for this module. **True:** Development error detection enabled **False:** Development error detection disabled

Table 4-7. Attribute WdgDevErrorDetect (WdgGeneral) detailed description

Property	Value
Label	Development Error Detection

Table continues on the next page...

Table 4-7. Attribute WdgDevErrorDetect (WdgGeneral) detailed description (continued)

Property	Value
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.5.2 WdgDisableAllowed (WdgGeneral)

Wdg Disable Allowed

Compile switch to allow / forbid disabling the watchdog driver during runtime. **True:** Disabling the watchdog driver at runtime is allowed **False:**Disabling the watchdog driver at runtime is not allowed

Table 4-8. Attribute WdgDisableAllowed (WdgGeneral) detailed description

Property	Value
Label	Wdg Disable Allowed
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.5.3 WdgMasterAccessProtection (WdgGeneral)

WdgMasterAccessProtection

Master Access Protection settings for the watchdog driver's.

Note

If use Core0 is necessary at least MAP0=1 and if use Core1 is necessary at least MAP1=1.

Table 4-9. Attribute WdgMasterAccessProtection (WdgGeneral) detailed description

Property	Value
Type	ENUMERATION
Origin	Custom

Table continues on the next page...

Table 4-9. Attribute WdgMasterAccessProtection (WdgGeneral) detailed description (continued)

Property	Value
Symbolic Name	false
Default	ALL_MAPx
Range	ALL_MAPx MAP0_MAP1 MAP0 MAP1 MAP2 MAP3 MAP4 MAP5 MAP6 MAP7

4.5.4 WdgIndex (WdgGeneral)

Wdg Index

Represents the watchdog driver's ID so that it can be referenced by the watchdog interface.

Table 4-10. Attribute WdgIndex (WdgGeneral) detailed description

Property	Value
Label	Wdg Index
Type	INTEGER
Origin	AUTOSAR_ECUC
Symbolic Name	true
Default	0
Invalid	Range <=255 >=0

4.5.5 WdgInitialTimeout (WdgGeneral)

Wdg Initial Timeout

The initial timeout (sec) for the trigger condition to be initialized during Init function. It shall be not larger than WdgMaxTimeout.

Table 4-11. Attribute WdgInitialTimeout (WdgGeneral) detailed description

Property	Value
Label	Wdg Initial Timeout [s]
Type	FLOAT
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	0.1
Invalid	Range <=65.535 >=0

4.5.6 WdgMaxTimeout (WdgGeneral)

Wdg Max Timeout

The maximum timeout (sec) to which the watchdog trigger condition can be initialized.

Table 4-12. Attribute WdgMaxTimeout (WdgGeneral) detailed description

Property	Value
Label	Wdg Max Timeout [s]
Type	FLOAT
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	0.2
Invalid	Range <=65.535 >=0

4.5.7 WdgRunArea (WdgGeneral)

Wdg Run Area

Represents the watchdog driver execution area is either from ROM(Flash) or RAM as required with the particular microcontroller.

Table 4-13. Attribute WdgRunArea (WdgGeneral) detailed description

Property	Value
Label	Wdg Run Area
Type	ENUMERATION

Table continues on the next page...

Table 4-13. Attribute WdgRunArea (WdgGeneral) detailed description (continued)

Property	Value
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	ROM
Range	RAM ROM

4.5.8 WdgCallbackNotification0 (WdgGeneral)

WdgCallbackNotification0

Callback notification for the ISR Wdg_Isr0 function

Table 4-14. Attribute WdgCallbackNotification0 (WdgGeneral) detailed description

Property	Value
Label	Wdg CallBack Notification 0
Type	FUNCTION-NAME
Origin	Custom
Symbolic Name	false
Default	NULL_PTR

4.5.9 WdgTriggerLocation (WdgGeneral)

Wdg Trigger Location

Location (memory address) of the watchdog trigger routine.

Note

Not supported by the current hardware.

Table 4-15. Attribute WdgTriggerLocation (WdgGeneral) detailed description

Property	Value
Label	Wdg Trigger Location
Type	FUNCTION-NAME
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	pointer

4.5.10 WdgVersionInfoApi (WdgGeneral)

Wdg VersionInfo Api

Compile switch to enable / disable the version information API. **True:** API enabled
False:API disabled

Table 4-16. Attribute WdgVersionInfoApi (WdgGeneral) detailed description

Property	Value
Label	Provide Version Info API
Type	BOOLEAN
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	true

4.6 Form WdgPublishedInformation

WdgPublishedInformation

Container holding all Wdg specific published information parameters

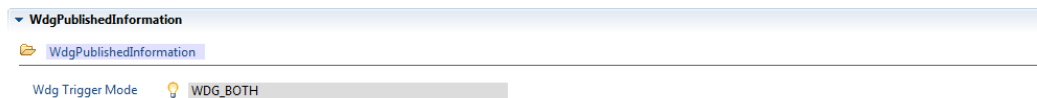


Figure 4-5. Tresos Plugin snapshot for WdgPublishedInformation form.

4.6.1 WdgTriggerMode (WdgPublishedInformation)

Wdg Trigger Mode

Watchdog trigger mode (toggle/window/both).

Table 4-17. Attribute WdgTriggerMode (WdgPublishedInformation) detailed description

Property	Value
Label	Wdg Trigger Mode
Type	ENUMERATION_LABEL
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	WDG_BOTH
Range	WDG_BOTH WDG_TOGGLE WDG_WINDOW

4.7 Form CommonPublishedInformation

CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

CommonPublishedInformation	
Name	Value
AUTOSAR Release Major Version	4
AUTOSAR Release Minor Version	0
AUTOSAR Release Revision Version	3
Numeric Module ID	102
Software Major Version	1
Software Minor Version	0
Software Patch Version	1
Vendor Api Infix	
Vendor ID	43

Figure 4-6. Tresos Plugin snapshot for CommonPublishedInformation form.

4.7.1 ArReleaseMajorVersion (CommonPublishedInformation)

AUTOSAR Release Major Version

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Table 4-18. Attribute ArReleaseMajorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	AUTOSAR Release Major Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	4
Invalid	Range >=4 <=4

4.7.2 ArReleaseMinorVersion (CommonPublishedInformation)

AUTOSAR Release Minor Version

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Table 4-19. Attribute ArReleaseMinorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	AUTOSAR Release Minor Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range >=0 <=0

4.7.3 ArReleaseRevisionVersion (CommonPublishedInformation)

AUTOSAR Patch Version

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

**Table 4-20. Attribute ArReleaseRevisionVersion
(CommonPublishedInformation) detailed description**

Property	Value
Label	AUTOSAR Release Revision Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	3
Invalid	Range >=3 <=3

4.7.4 ModuleId (CommonPublishedInformation)

Module ID

Module ID of this module from Module List.

Table 4-21. Attribute ModuleId (CommonPublishedInformation) detailed description

Property	Value
Label	Numeric Module ID
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	102
Invalid	Range >=102 <=102

4.7.5 SwMajorVersion (CommonPublishedInformation)

Software Major Version

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Table 4-22. Attribute SwMajorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Software Major Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	1
Invalid	Range >=1 <=1

4.7.6 SwMinorVersion (CommonPublishedInformation)

Software Minor Version

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Table 4-23. Attribute SwMinorVersion (CommonPublishedInformation) detailed description

Property	Value
Label	Software Minor Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range >=0 <=0

4.7.7 SwPatchVersion (CommonPublishedInformation)

Software Patch Version

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

**Table 4-24. Attribute SwPatchVersion (CommonPublishedInformation)
detailed description**

Property	Value
Label	Software Patch Version
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	1
Invalid	Range >=1 <=1

4.7.8 VendorApiInfix (CommonPublishedInformation)

Vendor Api Infix

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name. This parameter is used to specify the vendor specific name. In total, the implementation specific name is generated as follows:

<ModuleName>_>VendorId>_<VendorApiInfix><Api name from SWS>. E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name Can_Write defined in the SWS will translate to Can_123_v11r456Write. This parameter is mandatory for all modules with upper multiplicity > 1. It shall not be used for modules with upper multiplicity =1.

**Table 4-25. Attribute VendorApiInfix (CommonPublishedInformation)
detailed description**

Property	Value
Label	Vendor Api Infix
Type	STRING_LABEL
Origin	Custom
Symbolic Name	false
Default	
Enable	false

4.7.9 VendorId (CommonPublishedInformation)

Vendor ID

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Table 4-26. Attribute VendorId (CommonPublishedInformation) detailed description

Property	Value
Label	Vendor ID
Type	INTEGER_LABEL
Origin	Custom
Symbolic Name	false
Default	43
Invalid	Range >=43 <=43

4.8 Form WdgSettingsConfig

WdgSettingsConfig

Configuration items for the different watchdog settings, including those for external watchdog hardware.

The screenshot shows the 'General' tab of the 'WdgSettingsConfig_0' configuration window. The settings are as follows:

- Name:** WdgSettingsConfig_0
- Wdg Hardware Instance:** SWT0
- Wdg Default Mode:** WDGIF_SLOW_MODE
- WdgKeyedService:** [Icon]
- WdgServiceKeyValue (0 -> 65535):** 0
- Wdg External Trigger Counter:** /Gpt/Gpt/GptChannelConfigSet_0/GptChannelConfiguration_0
- WdgExternalConfiguration:**
 - Name:** WdgExternalConfiguration
 - WdgExternalContainerRef:** [Icon]
- WdgSettingsFast:**
 - Name:** WdgSettingsFast
 - Wdg Read Only:** [Icon]
 - Wdg Window Mode:** [Icon]
 - Wdg Runs In Debug Mode:** [Icon]
 - Wdg Operation Mode:** ResetOnTimeOut
 - Wdg Hard Lock:** [Icon]
 - Wdg Runs In Stop mode:** [Icon]

Figure 4-7. Tresos Plugin snapshot for WdgSettingsConfig form.

4.8.1 WdgInstance (WdgSettingsConfig)

Wdg Hardware Instance

Select Hardware Instance for watchdog driver initialization.

Table 4-27. Attribute WdgInstance (WdgSettingsConfig) detailed description

Property	Value
Label	Wdg Hardware Instance
Type	ENUMERATION
Origin	Custom
Symbolic Name	false

4.8.2 WdgDefaultMode (WdgSettingsConfig)

Wdg Default Mode

Default mode for watchdog driver initialization.

Table 4-28. Attribute WdgDefaultMode (WdgSettingsConfig) detailed description

Property	Value
Label	Wdg Default Mode
Type	ENUMERATION
Origin	AUTOSAR_ECUC
Symbolic Name	false
Default	WDGIF_SLOW_MODE
Range	WDGIF_FAST_MODE WDGIF_OFF_MODE WDGIF_SLOW_MODE

4.8.3 WdgKeyedService (WdgSettingsConfig)

Fixed Service Sequence or Keyed Service Mode

. False = Fixed Service Sequence, the fixed sequence 0xA602, 0xB480 is used to service the watchdog. True = Keyed Service Mode, two pseudorandom key values are used to service the watchdog.

Note

This is an Implementation Specific Parameter.

Table 4-29. Attribute WdgKeyedService (WdgSettingsConfig) detailed description

Property	Value
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.4 WdgServiceKeyValue (WdgSettingsConfig)

Service Key

This value is the initial service key value used in keyed service mode. If SWT_CR[KEY] is set, the next key value to be written to the SWT_SR is $(17 * SK + 3) \bmod 2^{16}$.

Table 4-30. Attribute WdgServiceKeyValue (WdgSettingsConfig) detailed description

Property	Value
Type	INTEGER
Origin	Custom
Symbolic Name	false
Default	0
Invalid	Range ../WdgKeyedService <=65535 >=0

4.8.5 WdgExternalTriggerCounterRef (WdgSettingsConfig)

Wdg External Trigger Counter

Reference to a Gpt channel for the Watchdog servicing routine implementation.

Note

Gpt channel for servicing WDG should be configured in continuous mode. Otherwise, WDG is always expired if a long time out is passed for Wdg_SetTriggerCondition.

Table 4-31. Attribute WdgExternalTriggerCounterRef (WdgSettingsConfig) detailed description

Property	Value
Label	Wdg External Trigger Counter
Type	CHOICE-REFERENCE
Origin	Custom

4.8.6 Form WdgSettingsFast

WdgSettingsFast

Hardware dependent settings for the watchdog driver's fast mode.

Is included by form : [Form WdgSettingsConfig](#)

The screenshot shows the 'WdgSettingsFast' configuration window. It has a title bar with a dropdown arrow and the text 'WdgSettingsFast'. Below the title bar is a tab labeled 'WdgSettingsFast'. The main area contains several settings, each with a small icon, a checkbox, and a dropdown menu:

- Wdg Read Only***: Icon of a document with a lock, checkbox is unchecked, dropdown is set to 'ResetOnTimeOut'.
- Wdg Window Mode***: Icon of a document with a lock, checkbox is unchecked, dropdown is set to 'BusError'.
- Wdg Runs In Debug Mode***: Icon of a document with a lock, checkbox is unchecked, dropdown is set to 'OscillatorClock'.
- Wdg Operation Mode***: Icon of a document with a lock, checkbox is unchecked, dropdown is set to 'ResetOnTimeOut'.
- Wdg Reset On Invalid Access***: Icon of a document with a lock, checkbox is unchecked, dropdown is set to 'BusError'.
- Wdg Clock Selection***: Icon of a document with a lock, checkbox is unchecked, dropdown is set to 'OscillatorClock'.
- Wdg Timeout Period [s] (0.002 -> 33554.432)***: Icon of a document with a lock, checkbox is unchecked, dropdown is set to '2.0'.
- Wdg Window Period[s] (0.002 -> 33554.432)***: Icon of a document with a lock, checkbox is unchecked, dropdown is set to '1.0'.
- Wdg Hard Lock***: Icon of a document with a lock, checkbox is unchecked, dropdown is set to 'Wdg Runs In Stop mode*'.
- Wdg Runs In Stop mode***: Icon of a document with a lock, checkbox is checked, dropdown is set to 'Wdg Runs In Stop mode*'.

Figure 4-8. Tresos Plugin snapshot for WdgSettingsFast form.

4.8.6.1 WdgReadOnly (WdgSettingsFast)

Wdg Read Only

This is the Implementation Specific parameter.Soft Lock for the Software Watchdog Timer Control (SWTCR) Register

- Enabled : SWTCR can be read or written.
- Disabled: SWTCR can be read only. A system reset is required before this register can again be written.

The setting of this switch is intended to prevent accidental writes of the SWTCR from changing the defined system watchdog configuration.

Table 4-32. Attribute WdgReadOnly (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Read Only
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.6.2 WdgHardLock (WdgSettingsFast)

Wdg Hard Lock.

- Disabled: Hard Lock is not set.
- Enabled : Hard Lock is set.

Table 4-33. Attribute WdgHardLock (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Hard Lock
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.6.3 WdgWindowMode (WdgSettingsFast)

Wdg WindowMode.

- Disabled: Regular mode, service sequence can be done at any time.
- Enabled : Windowed mode, the service sequence is only valid when the down counter is less than value in the SWTWN register.

Table 4-34. Attribute WdgWindowMode (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Window Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.6.4 WdgRunsInStopMode (WdgSettingsFast)

Wdg Runs in Stop Mode

This is the Implementation Specific parameter.

- Enabled: SWT continues to count even while the processor core is in stop mode.
- Disabled: SWT stops counting if the processor core is in stop mode.

Note

The 'WdgRunsInStopMode' parameter specifies if the watchdog timer should run or not while the clock to the core is halted. This is true only for the STOP0 mode of the controller. It will always run while the controller is in the HALT0 mode.

Table 4-35. Attribute WdgRunsInStopMode (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Runs In Stop mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.8.6.5 WdgRunsInDebugMode (WdgSettingsFast)

Wdg Runs In Debug Mode

This is the Implementation Specific parameter.

- Enabled: SWT continues to count even while the device enters the debug mode.
- Disabled: SWT stops counting if the processor core when the device enters the debug mode

Table 4-36. Attribute WdgRunsInDebugMode (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Runs In Debug Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.6.6 WdgOperationMode (WdgSettingsFast)

Wdg Operation Mode

This is the Implementation Specific parameter.

- ResetOnTimeOut: Generate a reset on a time-out.
- Interrupt: Generate an interrupt on an initial time-out, reset on a second consecutive time-out.

Table 4-37. Attribute WdgOperationMode (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Operation Mode
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	ResetOnTimeOut
Range	ResetOnTimeOut Interrupt

4.8.6.7 WdgResetOnInvalidAccess (WdgSettingsFast)

Wdg Reset On Invalid Access

If window mode is enabled, the service sequence must be performed in the last part of the window time out period. The window is open when the down counter is less than the value in the SWT_WN register. Outside of this window, service sequence writes are invalid access and generate:

- BUS error: Invalid access to the SWT generated a bus error
- System Reset: Invalid access to the SWT causes a system reset (if watchdog is enabled).

Table 4-38. Attribute WdgResetOnInvalidAccess (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Reset On Invalid Access
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	BusError
Range	BusError SystemReset

4.8.6.8 WdgClockSelection (WdgSettingsFast)

Wdg Clock Selection

The unique SWT counter clock is the undivided low power internal oscillator (IRC128Khz).

Table 4-39. Attribute WdgClockSelection (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Clock Selection
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	OscillatorClock
Range	OscillatorClock SystemClock

4.8.6.9 WdgTimeoutPeriod (WdgSettingsFast)

Wdg Timeout Period

This is the Implementation Specific parameter. Software Watchdog Time-Out Period in seconds. Selects the time-out period for the SWT.

Table 4-40. Attribute WdgTimeoutPeriod (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Timeout Period [s]
Type	FLOAT
Origin	Custom
Symbolic Name	false
Default	2
Invalid	Range <=33554.432 >=0.002

4.8.6.10 WdgWindowPeriod (WdgSettingsFast)

Wdg Window Period

This is the Implementation Specific parameter. Window start value. When window mode is enabled, the service sequence can only be written when the internal down counter is less than this value.

Table 4-41. Attribute WdgWindowPeriod (WdgSettingsFast) detailed description

Property	Value
Label	Wdg Window Period[s]
Type	FLOAT
Origin	Custom
Symbolic Name	false
Default	1
Invalid	Range <=33554.432 >=0.002

4.8.7 Form WdgSettingsOff

WdgSettingsOff

Hardware dependent settings for the watchdog driver's off mode.

Is included by form : [Form WdgSettingsConfig](#)

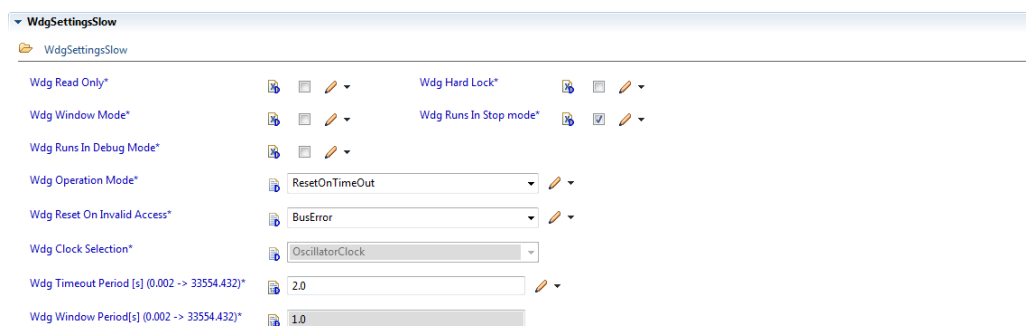


Figure 4-9. Tresos Plugin snapshot for WdgSettingsOff form.

4.8.7.1 WdgReadOnly (WdgSettingsOff)

WdgReadOnly

This is the Implementation Specific parameter.Soft Lock for the Software Watchdog Timer Control (SWTCR) Register

- Enabled : SWTCR can be read or written.
- Disabled: SWTCR can be read only. A system reset is required before this register can again be written.

The setting of this switch is intended to prevent accidental writes of the SWTCR from changing the defined system watchdog configuration.

Table 4-42. Attribute WdgReadOnly (WdgSettingsOff) detailed description

Property	Value
Label	Wdg Read Only
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.7.2 WdgHardLock (WdgSettingsOff)

Wdg Hard Lock.

- Disabled: Hard Lock is not set.
- Enabled : Hard Lock is set.

Table 4-43. Attribute WdgHardLock (WdgSettingsOff) detailed description

Property	Value
Label	Wdg Hard Lock
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.8 Form WdgSettingsSlow

WdgSettingsSlow

Hardware dependent settings for the watchdog driver's slow mode.

Is included by form : [Form WdgSettingsConfig](#)

Figure 4-10. Tresos Plugin snapshot for WdgSettingsSlow form.

4.8.8.1 WdgReadOnly (WdgSettingsSlow)

Wdg Read Only

This is the Implementation Specific parameter.Soft Lock for the Software Watchdog Timer Control (SWTCR) Register

- Enabled : SWTCR can be read or written.
- Disabled: SWTCR can be read only. A system reset is required before this register can again be written.

The setting of this switch is intended to prevent accidental writes of the SWTCR from changing the defined system watchdog configuration.

Table 4-44. Attribute WdgReadOnly (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Read Only
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.8.2 WdgHardLock (WdgSettingsSlow)

Wdg Hard Lock.

- Disabled: Hard Lock is not set.
- Enabled : Hard Lock is set.

Table 4-45. Attribute WdgHardLock (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Hard Lock
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.8.3 WdgWindowMode (WdgSettingsSlow)

Wdg WindowMode

- Disabled: Regular mode, service sequence can be done at any time.
- Enabled : Windowed mode, the service sequence is only valid when the down counter is less than value in the SWTWN register.

Table 4-46. Attribute WdgWindowMode (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Window Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.8.4 WdgRunsInStopMode (WdgSettingsSlow)

Wdg Runs in Stop Mode

This is the Implementation Specific parameter.

- Enabled: SWT continues to count even while the processor core is in stop mode.
- Disabled: SWT stops counting if the processor core is in stop mode.

Note

The 'WdgRunsInStopMode' parameter specifies if the watchdog timer should run or not while the clock to the core is halted. This is true only for the STOP0 mode of the controller. It will always run while the controller is in the HALT0 mode.

Table 4-47. Attribute WdgRunsInStopMode (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Runs In Stop mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	true

4.8.8.5 WdgRunsInDebugMode (WdgSettingsSlow)

Wdg Runs In Debug Mode

This is the Implementation Specific parameter.

- Enabled: SWT continues to count even while the device enters the debug mode.
- Disabled: SWT stops counting if the processor core when the device enters the debug mode.

Table 4-48. Attribute WdgRunsInDebugMode (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Runs In Debug Mode
Type	BOOLEAN
Origin	Custom
Symbolic Name	false
Default	false

4.8.8.6 WdgOperationMode (WdgSettingsSlow)

Wdg Operation Mode

This is the Implementation Specific parameter.

- ResetOnTimeOut: Generate a reset on a time-out.
- Interrupt: Generate an interrupt on an initial time-out, reset on a second consecutive time-out.

Table 4-49. Attribute WdgOperationMode (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Operation Mode
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	ResetOnTimeOut
Range	ResetOnTimeOut Interrupt

4.8.8.7 WdgResetOnInvalidAccess (WdgSettingsSlow)

Wdg Reset On Invalid Acces

If window mode is enabled, the service sequence must be performed in the last part of the window time out period. The window is open when the down counter is less than the value in the SWT_WN register. Outside of this window, service sequence writes are invalid access and generate:

- BUS error: Invalid access to the SWT generated a bus error
- System Reset: Invalid access to the SWT causes a system reset (if wathdog is enabled).

Table 4-50. Attribute WdgResetOnInvalidAccess (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Reset On Invalid Access
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	BusError
Range	BusError SystemReset

4.8.8.8 WdgClockSelection (WdgSettingsSlow)

Wdg Clock Selection

The unique SWT counter clock is the undivided low power internal oscillator (IRC128Khz).

Table 4-51. Attribute WdgClockSelection (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Clock Selection
Type	ENUMERATION
Origin	Custom
Symbolic Name	false
Default	OscillatorClock
Range	OscillatorClock SystemClock

4.8.8.9 WdgTimeoutPeriod (WdgSettingsSlow)

Wdg Timeout Period

This is the Implementation Specific parameter. Software Watchdog Time-Out Period in seconds. Selects the time-out period for the SWT.

Table 4-52. Attribute WdgTimeoutPeriod (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Timeout Period [s]
Type	FLOAT
Origin	Custom
Symbolic Name	false
Default	2
Invalid	Range <=33554.432 >=0.002

4.8.8.10 WdgWindowPeriod (WdgSettingsSlow)

Wdg Window Period

This is the Implementation Specific parameter. Window start value. When window mode is enabled, the service sequence can only be written when the internal down counter is less than this value.

Table 4-53. Attribute WdgWindowPeriod (WdgSettingsSlow) detailed description

Property	Value
Label	Wdg Window Period[s]
Type	FLOAT
Origin	Custom
Symbolic Name	false
Default	1
Invalid	Range <=33554.432 >=0.002

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